BASE Properties

ACID vs BASE

ACID

- Atomicity
- Consistency
- Isolation
- Durability

BASE

- Basically Available
  - Robust to cluster node failure
- Soft state
  - State can change over time, even w/o input, because …
- Eventual consistency
  - Not immediate not always

ACID vs BASE

ACID

- Pessimistic
- Cannot tolerate inconsistency

- Force consistency now

BASE

- Optimistic
- Can tolerate some inconsistency

- Work toward consistency

Basically available

- Robust in the face of node failures
- Partial failures tolerated
  - Some data might not be available for queries
  - Often replicate data to mitigate

Soft state

- State can change over time, even w/o input, to allow eventual consistency
  - In the meantime, can be inconsistent

Eventually consistent

- If no input, eventually becomes consistent
  - Working towards consistency
- If input, cannot guaranteed consistency
  - Progress towards consistency might not catch up

- Reality: ops guaranteeing consistency are slow

Causal consistency

1. Monotonic read
   - Once you see a result, you never see an earlier value
2. Monotonic write
   - Updates are done in the order executed
3. Read-your-writes
   - If you write a record, you always see those updates.
4. Writes-follow-reads
   - Any reads before a write are done before the write

Conflict resolution

When? How?

- Must have policy.
- Typically on writes – e.g., last write wins

“Base: An Acid Alternative”

https://queue.acm.org/detail.cfm?id=1394128

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